SQRT: FAR Presentation



Presented by Bruce, Caimin, Elyana, Javi & Orlaith.

15-Apr-26

Key Elements of Mission & System Overview

Mission Statement: The SQRT mission will produce and observe the flash evaporation of water via exposure to a low-pressure and low-temperature environment to investigate how such plume-like vapour structures thermally evolve.

Mission Motivation

Space Based;
Icy Moon Plume Understanding

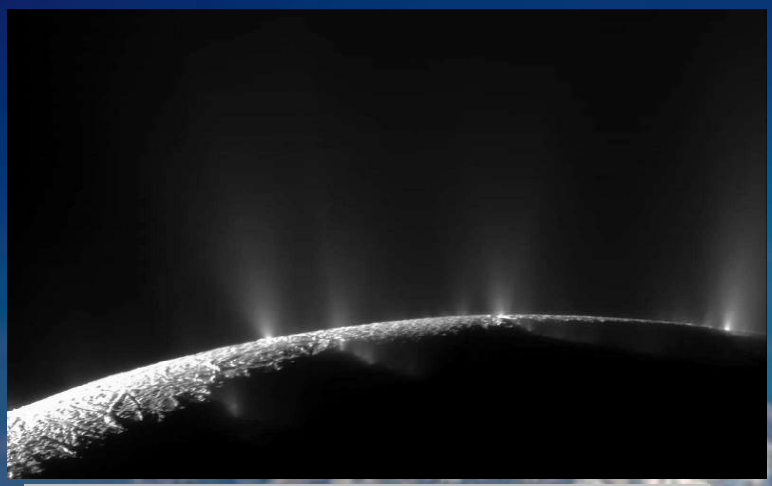


Image credit; NASA/JPL/SSI - https://www.space.com/30944-nasa-cassini-saturn-moon-enceladus-flyby.html

Earth Based;
Desalination Processes Understanding



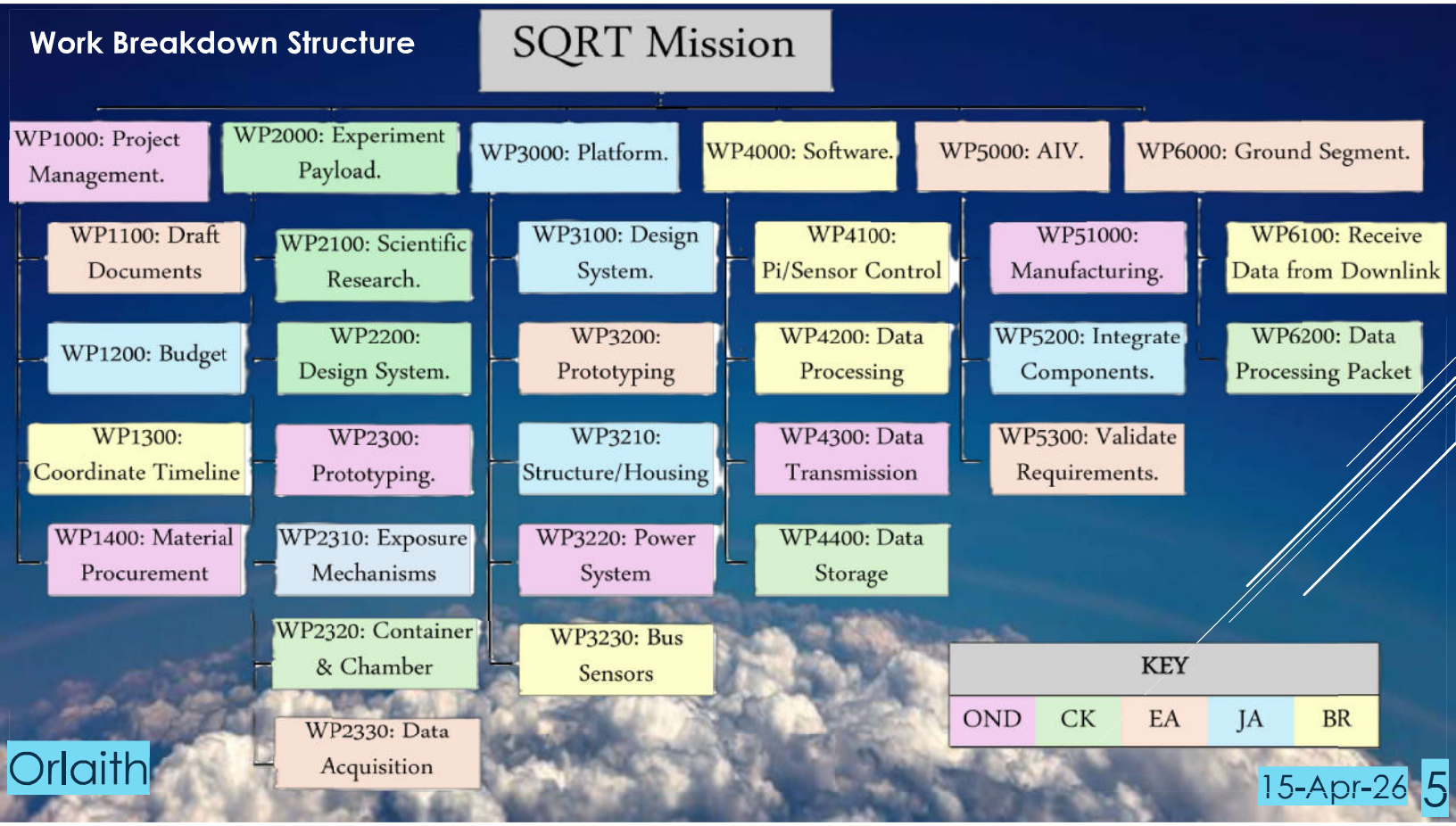
Image credit: https://condorchem.com/en/blog/evaporation-systems-water-desalination/

Team Roles

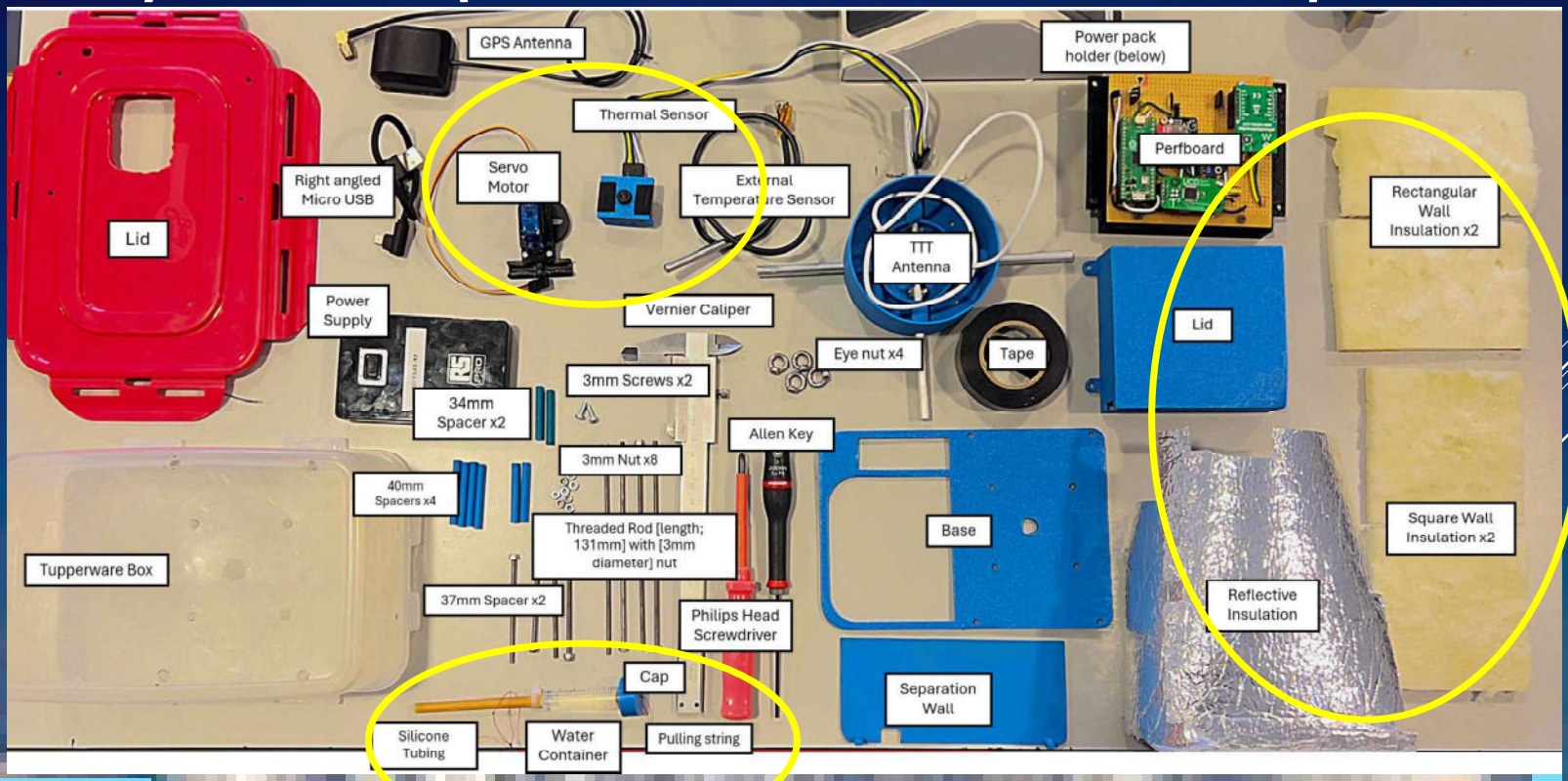
Team Member	Lead Role	Most Recent Major Contributions
Orlaith	Project Manager	Documentation, Hardware, Testing
Elyana	AIV	Testing & Verification, Documentation, Hardware
Javi	Structure	Hardware, Testing, CAD
Caimin	Science	Software, Electronics, Testing
Bruce	Software	Electronics, Software, Testing

Created by Orlaith, Presented by All

Work Breakdown Structure



Payload Components & What makes SQRT special



Core Experimental Components & Placement Purpose

Experimental Component	Placement Location	Purpose & Location Reasoning
Thermal Sensor	Wide angled view in low corner	To capture the thermal evolution of the water evaporation
Syringe/ Water Container	Directly below power pack and two tiers below electronics	To store water sample at appropriate temperature before release into experimental chamber
Servo Motor	Near water container and mounted around the syringe tubing	Programmed to turn at specific angle, thereby releasing water sample through tubing valve due to pressure difference
Water Container Cap and Pulling String	At syringe open end, covers rubber stopper, faced opposite from experimental chamber	Twistable, removable and strong force string to allow repeat testing. Assures no water leakage within electronics bus.
Multiple Insulating materials	Surrounding the electrics bus, syringe and tubing	To ensure the water stays in liquid form and reduces possibility of payload freeze.
Electronics Cap	Directly above electronics configuration on perfboard	To ensure that any potential leakage or moisture does not interact with electronics.

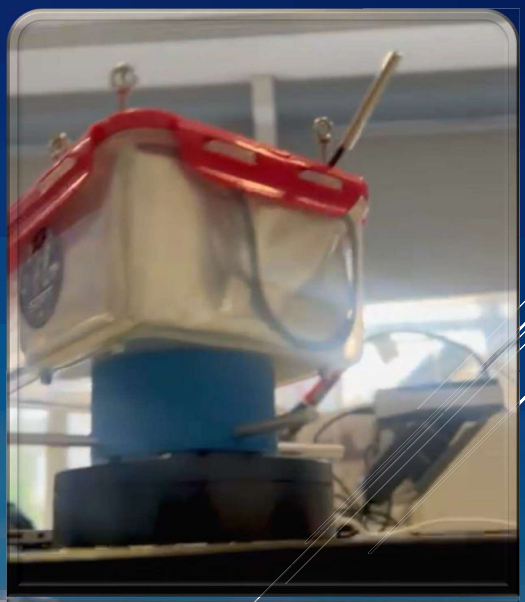
Payload Exterior View - Fully Assembled



Top Angled View



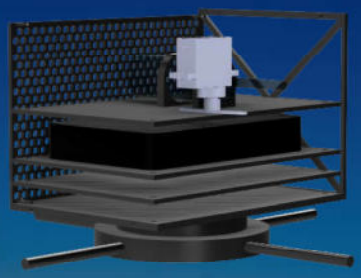
Side View

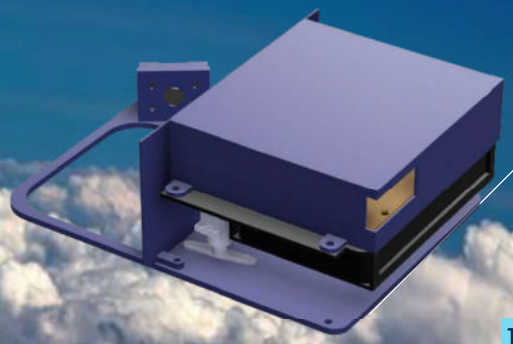


Bottom Angled View

Hardware – changes since CDR: Structure

Full Redesign:
- Unnecesarry components
  - lateral faces
  - Top
- Modified components
  - Base
  - Internal floors
- New components
  - Camera mounting
  - Separation wall
  - Electronics cap

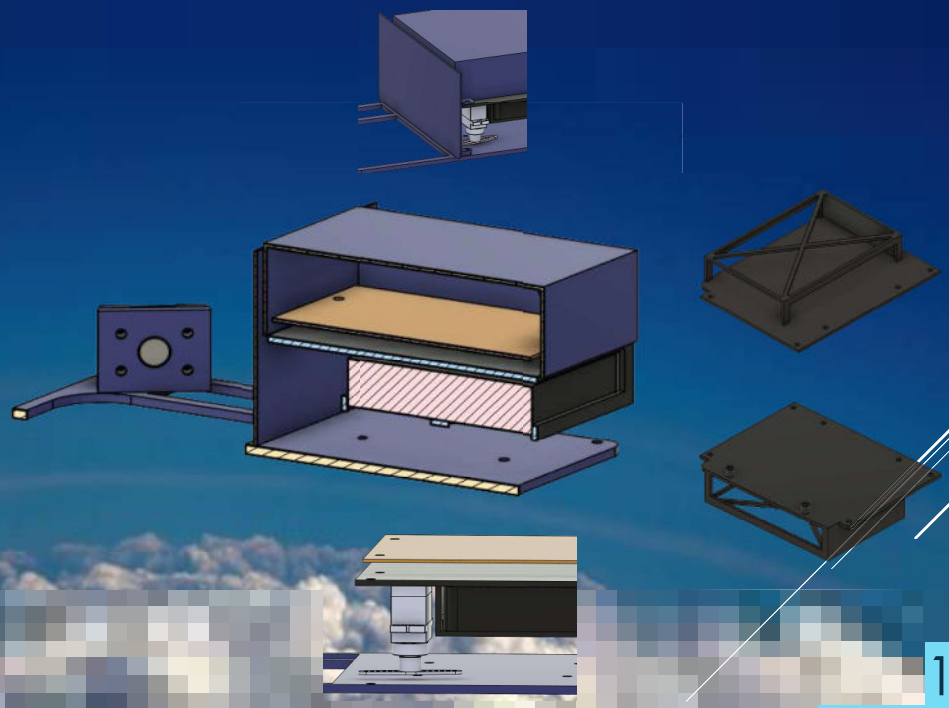




Javi

Hardware – changes since CDR: Structure

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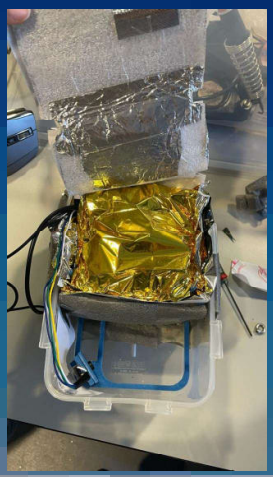
Javi

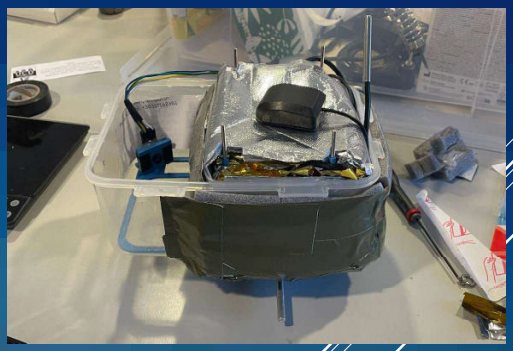
Hardware – changes since CDR: Thermal insulation

Thermal insulation is based on two process:

Passive shielding: three layers of aluminum-based material. One external layer of EPE foam sealed with tape

Acvite heating: aside from the battery and electronics, there is a heating pad in the elcetroncs bus near the syringe





Hardware – changes since CDR: Mass & volume

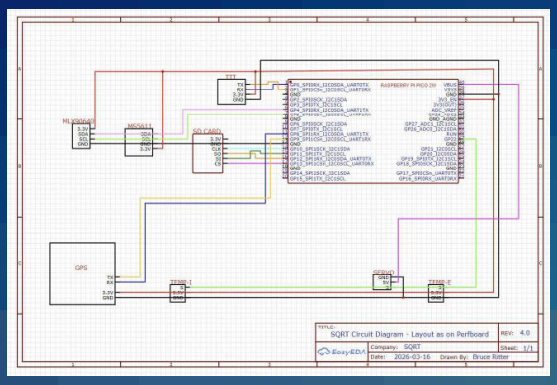
Need of active heating and extra thermal insulation -> **heavier**

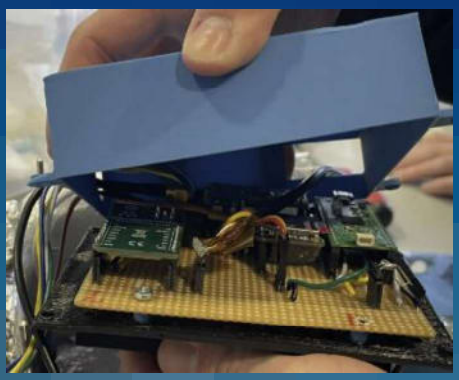
As two compartments were needed, the tupperwares under 2L did not meet the needs. **A 2.3L one is used**

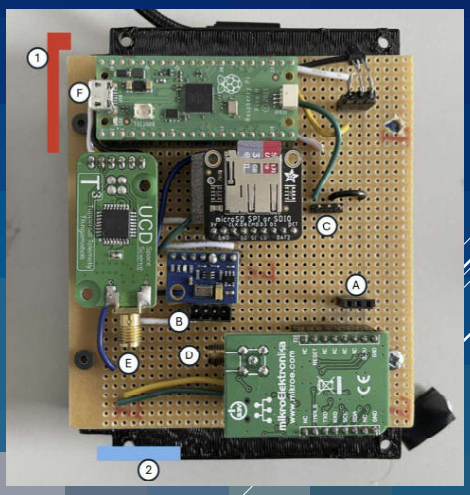


Javi

Electronics – changes since CDR



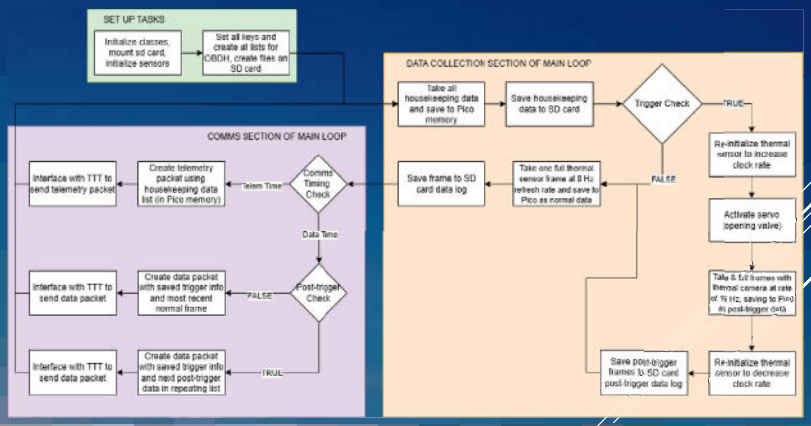




Bruce

Software – overview and status – changes since CDR

Component of Software	CDR Design	Final Software Product
OBDH	Data saved to and taken from SD card for triggering algorithm and comms Individual files for sensors	All data passed to triggering algorithm and comms comes from Pico memory One HK file, inclusion of error logging
Data Transmission	Had one version of data packet	Two data packet versions: pre and post trigger
Thermal Sensor Operation	Using 32 FPS (Less than quoted maximum)	Using 16 FPS post-trigger (Actual maximum)



Thermal Sensor Deviation

RFD Submitted for reduction in frame rate (to at least 8 fps)



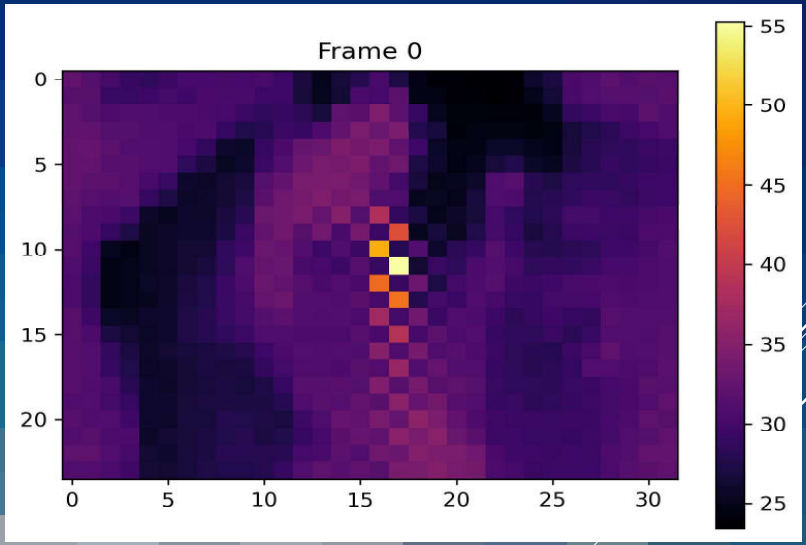
Expected	Real	Explanation
64 FPS maximum, can run at 32	16 FPS maximum	Highest achievable rate, achieved with I2C clock alterations
Would capture > 16 frames during flash evaporation	Will capture 3-4 frames during flash evaporation	Highest amount achievable with max FPS



Bruce

Thermal Sensor Reading Patterns

- Two modes available:
  Chess pattern mode used
- Alternating pixels are updated at each refresh point
- Frame rate per pixel is essentially halved
- Artifacts of this reading pattern can be seen when dynamic targets are captured (see figure adjacent)
- **Solution: A full frame is produced by combining consecutive frames.**

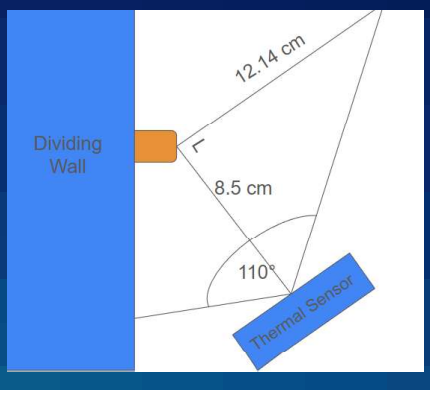


Caimin

Thermal Sensor Calibration

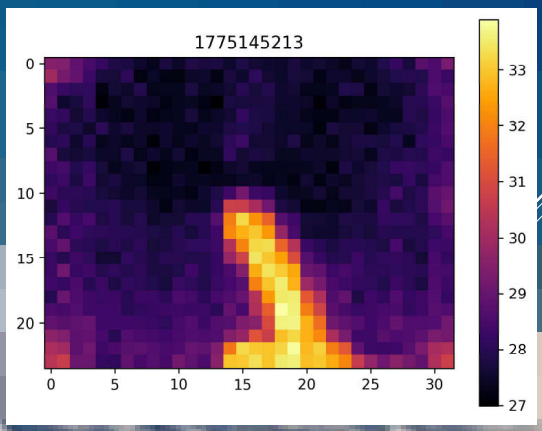
Spatial Resolution (~24x24 cm^2 field at ~ 8.5 cm distance)

- X-axis: ~ 0.75 cm per pixel
- Y-axis: ~ 1 cm per pixel



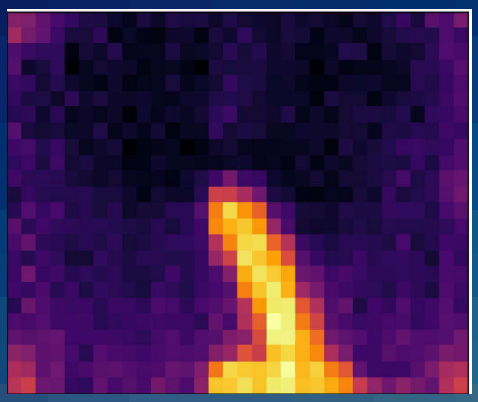
Science Target Positioning

- Estimated to be at end of finger in adjacent figure
- Onboard Cropping for transmission based on this position

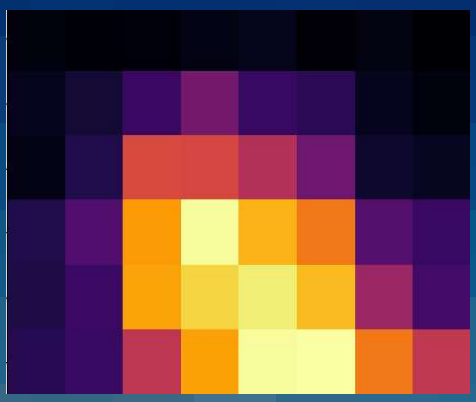


Caimin

Science Data Reduction



Original Thermal Array



Cropped Thermal Array

- Reductions in thermal sensor data transmitted are required to avoid UART buffer overflows
- Cropping method chosen to prevent loss of resolution
- Frame size reduced by a factor of 16
- Temperatures transmitted in hex format

Post-Mission Analysis

Onboard Storage

- SD card will be carefully removed from recovered TupperSat
- All stored data will be processed to provide each team member with a uniform datapack
- Full thermal frames (32x24)
- Comprehensive dataset

Downlinked Data

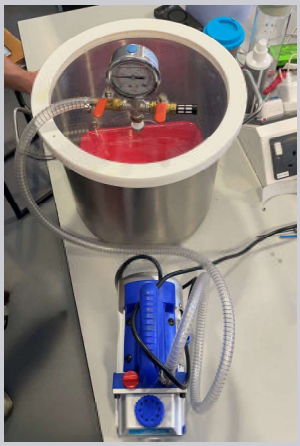
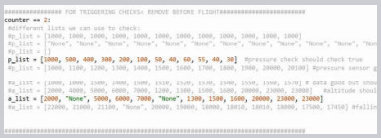
- Received data packets will be stored on groundstation device
- Reduced thermal frames (8x6)
- Reduced temporal resolution for all housekeeping/science data

Separate workflows written for each mode of processing (variations based on inputted file type/shape)

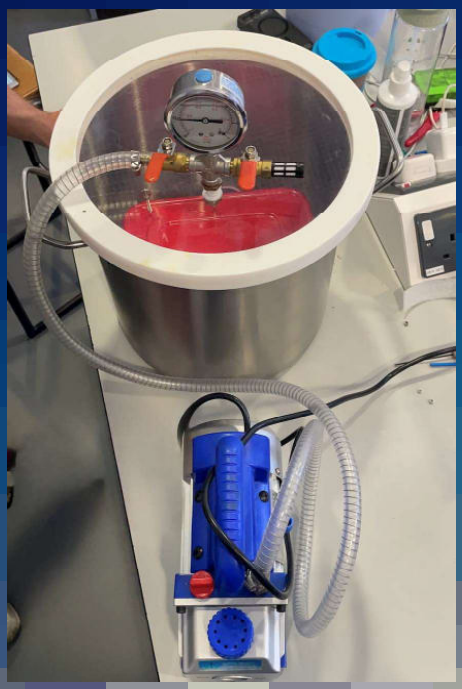
Workflow used will depend on TupperSat recovery

Key Tests

We performed testing at different levels:

Subsystem Test	Environmental Test	Software Tests	System Tests
Ambient Pressure Simulation: 	Insulation/Thermal Freezer Test: 	Triggering Algorithm Bench Test:  GPS Live Test: 	- Long Range Communication - 6H Endurance Test

Ambient Pressure Simulation



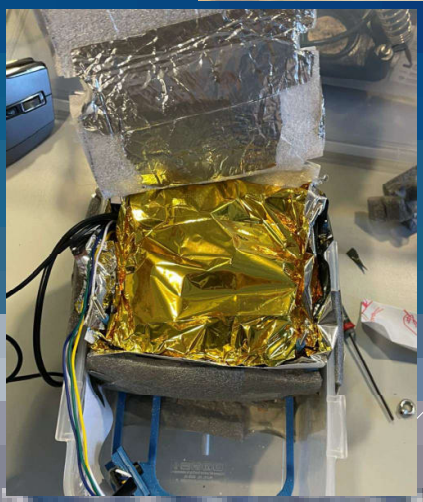
- Ensuring **structural integrity** of the water container under low pressure
- Verifying **no leakage** before valve activation
- Validating **triggering algorithm response** at pressure threshold
- Confirming **successful water ejection** after valve activation

Elyana

Thermal (Fridge) Test

- Ensuring that temperature differential was maintained between electronics bus and low-pressure chamber
- Verifying that water sample didn't freeze
- Multiple tests to review different insulation strategies and their respective efficiencies

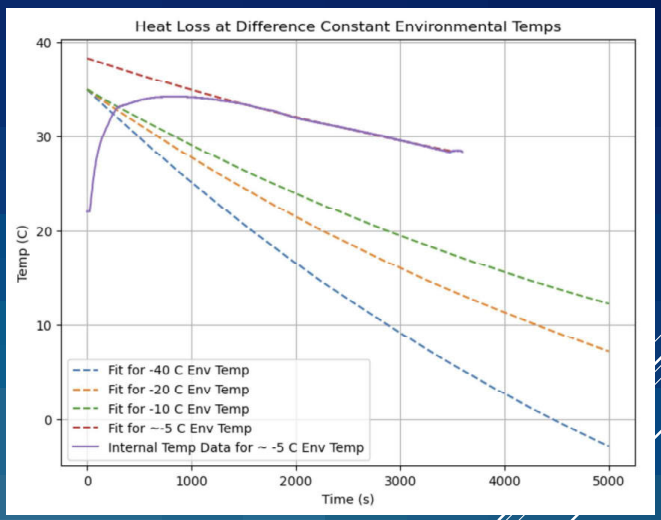


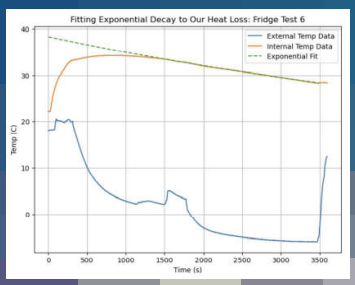


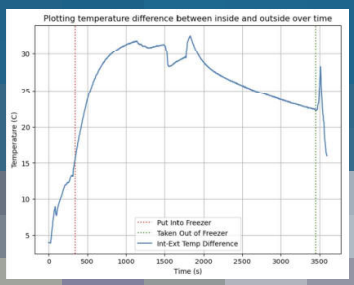
Elyana

Thermal (Fridge) Test

- Final test result: Temperature difference > 30 deg C with one heating pad
- Using an exponential fit we estimate a lower bound of 75 minutes before freezing (Given external temp is –40 deg C the entire time, air pressure ~1 bar)

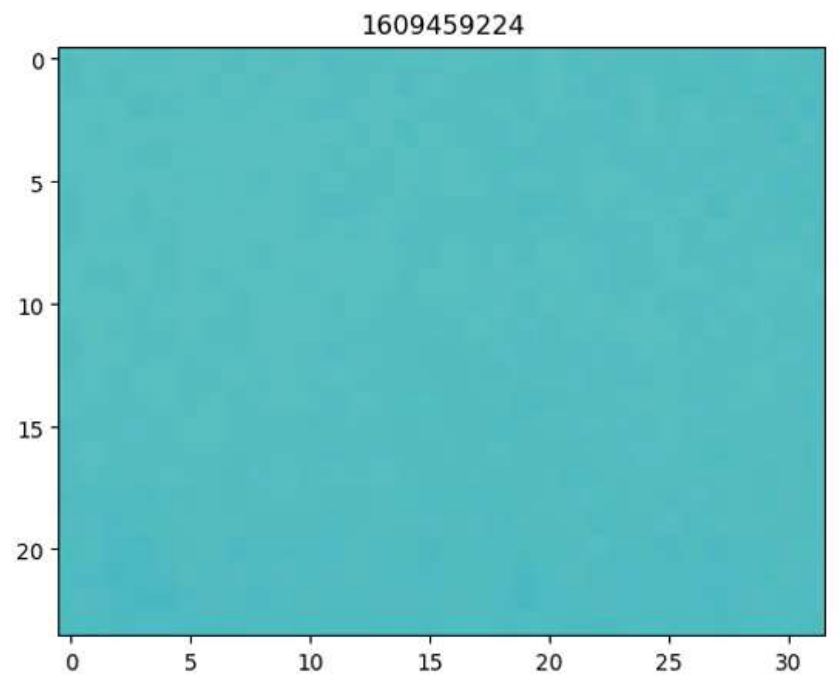






Elyana

Thermal Test 3



Frames from the third thermal test

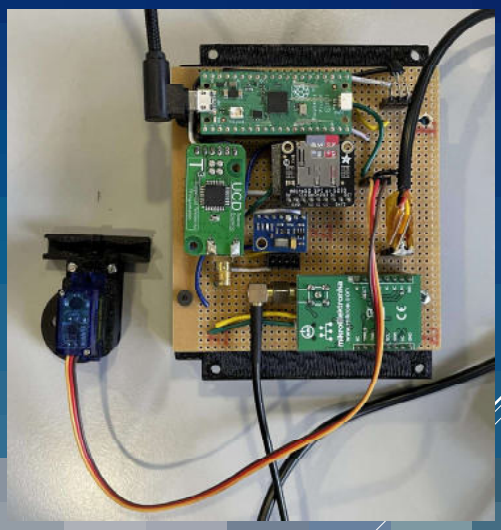
Variation in temperature can be observed from blue (warmer) to yellow/green (colder)

Dividing wall to electronics can be seen to maintain temperature

Triggering Algorithm Bench Test

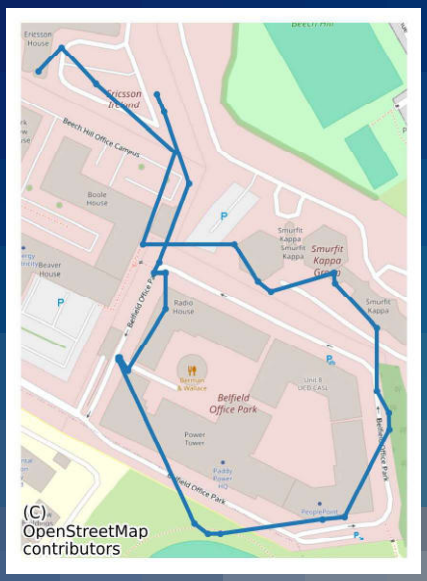
Requirement	Scenarios Required	Scenarios Passed	Pass (Y/N)
Req 1. SQT:TR:00700 Pressure Trigger	1, 4, 7	1, 4, 7	Y
Req 2. SQT:TR:00710 Altitude Trigger	2, 5, 7	2, 5, 7	Y
Req 3. SQT:TR:00720 Falling Trigger Operation	3, 6	3, 6	Y
Req 4. SQT:TR:00730 Falling Trigger Minimum Altitude	4, 8	4, 8	Y

Scenario	Screenshot of Ground Station Received Data	Screenshot of SD Data	Servo Arm	Pass (Y/N)
1	0|True|G|30|23000|	FALSE None 1017.231 None TRUE G 30 23000	90 deg	Y
2	0|True|B|1000|23000|	FALSE None 1017.186 None TRUE B 1000 23000	90 deg	Y
3	|True|U|50|17450|	FALSE None 1017.277 None TRUE U 50 17450	90 deg	Y
4	|False|None|1017.213|None|	FALSE None 1017.189 None FALSE None 50 1570	0 deg	Y
5	|True|B|None|23000|	FALSE None 1017.252 None TRUE B None 23000	90 deg	Y
6	0|True|U|None|17450|	FALSE None 1017.227 None TRUE U None 17450	90 deg	Y
7	|False|None|1017.241|None|	FALSE None 1017.188 None FALSE None 50 23000	0 deg	Y
8	0|False|None|1017.203|None	FALSE None 50 1200	0 deg	Y

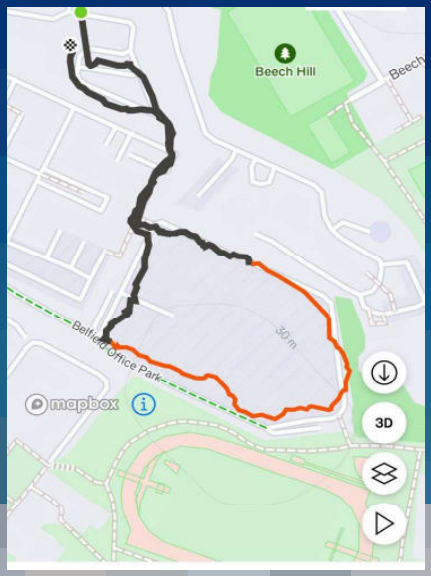


GPS Test

SQRT GPS Groundtrack



Strava Groundtrack



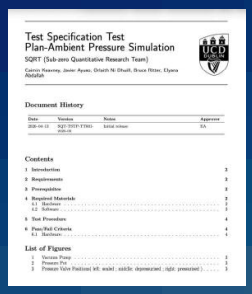
- Completed Earlier Today
- Maximum location deviation of 40 metres
- Altitude figures from GPS range from 20.1 metres to 64.9 metres (both within actual mean altitude of ~30 metres).

System Test

Long Range Communication Test (02/04)	6H Endurance Test (13/04)
Objective: **Validate communication over increasing distance** **Method:** • Continuous telemetry transmission • Increasing separation from ground station **Results:** • Stable telemetry transmission initially • No anomalies in transmitted data • Ground station failure interrupted test **Conclusion:** ➡️ **Communication system validated (issue external to payload)**	**Objective:** **Validate full system operation over mission duration** **Method:** • Continuous operation for 6 hours • Monitoring telemetry, sensors, battery **Results:** • Continuous operation for 6 hours • Stable telemetry and sensor data • Successful valve activation during test • No system interruptions observed **Conclusion:** ➡️ **System meets endurance requirements**

Verification and Test Documentation

VCD


Test Specification Test Plan-Ambient Pressure Simulation

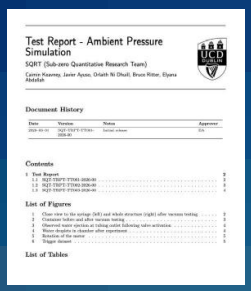
SQRT (Sub-zero Quantitative Research Team)

Document History

Contents

List of Figures

TSTP


Test Report - Ambient Pressure Simulation

SQRT (Sub-zero Quantitative Research Team)

Document History

Contents

List of Figures

List of Tables

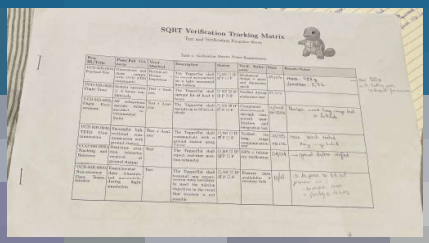
TRPT

Table 13: Verification Matrix: Triggering Algorithms Requirements

Requirement ID/Title	Verification Level	Verification Method	Test Reference	Pass/Fail Criteria	Status	Notes
SQT:TR:00700 Pressure Trigger	System + Subsystem	Test + Review-of-Design	SQT-TRTP-TT004- 2026-00 + SQT-TRPT-TT001-2026-00	Trigger activates when measured pressure ≤35 mbar	Pass	Verified via bench test using simulated telemetry data and via Ambient Pressure Simulation (for an adjusted pressure if 20mbar)
SQT:TR:00710 Altitude Trigger	Subsystem	Test + Analysis	SQT-TRTP-TT004- 2026-000	Trigger activates at altitude ≥ 23 km when pressure data flagged inconsistent	Pass	Verified via bench test using simulated telemetry data

Continued on next page

Verification Tracking Matrix

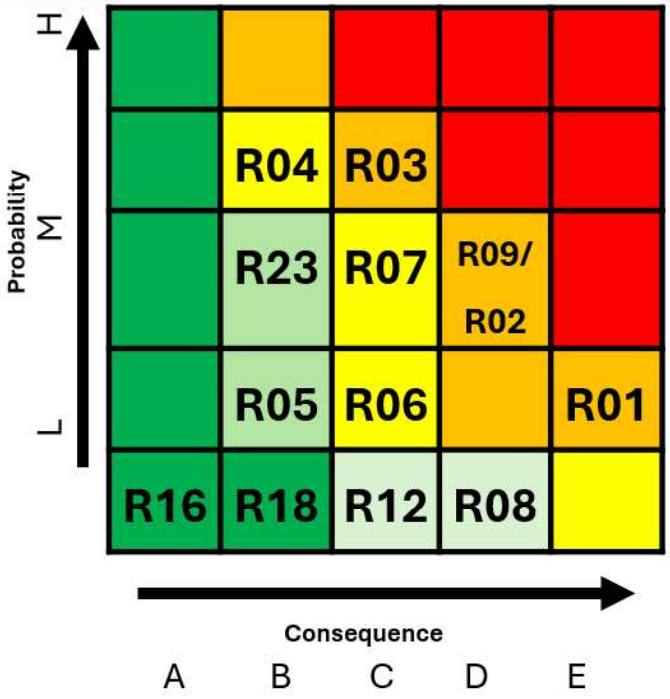

SQRT Verification Tracking Matrix

Name	Reqs Satisfied	P/F
Insulation Test	SQT:TR:00230 Tubing Material	P
Insulation Test	SQT:TR:00100 Water Temperature	P
Insulation Test	SQT:TR:00310 Sensor Function	P
Vacuum Trigger Test	SQT:TR:00700	P
Leak Avoidance Test	SQT:TR:00210	P
GPS Test	UCD:MR:0020	P
6H Endurance Test	UCD:MR:0020	P
Long Range Test	UCD:TR:00110	P

Plan for launch & Special handling required

- Keep payload inside a car before it is time to attach to the balloon (to help with heat)
- Avoid unnecessary movement or vibration during transport to prevent any premature water leakage or payload breakage.
- Ensure the payload is transported in a safe protective container.
- Avoid placement of payload container under any other object to avoid breakage due to pressure.
- Ensure electronics are powered ‘off’ throughout transport journey to avoid battery loss.
- Ensure the eye-nuts are sufficiently tightened after connection to balloon.

Risks & Mitigation



R01. Water freezing prematurely
R02. Valve failure
R03. Low-pressure chamber environment failure
R09. Vents/valve/tubing freezing & plugging during exposure
R04. Electrical component failure
R06. Balloon fails to reach necessary height
R07. Motor failure
R18. Structural damage from testing before launch
R05. Data storage failure
R08. Insufficient pressure drop within payload
R12. Pressure differential structural stress
R23. Payload cannot be retrieved
R16. Condensation on camera lens

Final Readiness Statement & Conclusions

Formal Declaration of Mission Readiness;

- The SQRT payload has completed design, integration, and verification testing.
- All required documentation and subsystem checks (EXCEPT THE Pico memory) are complete, and remaining risks are acceptable.
- The team considers the payload ready for flight, pending final launch-day GO/NO-GO checks.

Thank you for your attention…
Any questions?

